Over The Air Computing
Master Thesis

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Abstract **– Over-the-Air (OTA) is an emerging technique that leverages the superposition property of wireless channels to perform computations directly during the transmission process. This approach has the potential to significantly reduce latency and energy consumption in distributed systems, making it particularly suitable for applications in Internet of Things (Internet of Things (IoT)) networks, edge computing, and real-time data processing. This thesis explores the theoretical foundations of OTA, including channel modeling, signal processing techniques, and error analysis. Practical implementations and case studies demonstrating the effectiveness of OTA in various scenarios are also presented.**

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List of Acronyms and Abbreviations

AWGN	Additive White Gaussian Noise 3–6
CSI	Channel State Information 3, 8
ED	End Device 3, 5, 7, 8
ES	Edge Server 5
FC	Fusion Center 3, 5, 7, 8
FIR	Finite Impulse Response 5
IoT	Internet of Things 1
ISI	Inter-Symbol Interference 3, 8–10
MAC	Multiple Access Channel 3–5
MIMO	Multiple-Input Multiple-Output 11
MSE	Mean Squared Error 5, 8
OTA	Over-the-Air 1, 3, 5, 7, 9
PAM	Pulse Amplitude Modulation 5, 7
SDR	Software Defined Radio 3, 5
TDMA	Time-Division Multiple Access 11

Introduction

Over the Air Computing offers a paradigm shift in how distributed computations are performed in wireless networks. By exploiting the inherent properties of the wireless medium, OTA enables simultaneous transmissions from multiple devices to be combined in a way that directly computes a desired function of the transmitted data. This technique has garnered significant attention due to its potential to enhance efficiency and reduce latency in applications such as federated learning, distributed sensing, and real-time analytics. The motivation behind this thesis is to explore the practical implementation of OTA using Software Defined Radios Software Defined Radio (SDR) and MATLAB, addressing the challenges associated with synchronization, channel impairments, and waveform design. The research objectives include designing a prototype OTA network, validating its performance, and analyzing the impact of various impairments on computation accuracy.

0.a Background and Motivation

The increasing need for efficiency in computation-oriented applications where the interest lies in a function of local information, rather than the information itself [6]. The conventional approach of separating communication and computation often proves suboptimal [3], [4], [6]. The potential of OTA Computation is to achieve significantly higher computation rates by harnessing interference through signal superposition in a wireless Multiple Access Channel (MAC) [6].

0.b Problem Statement and Research Objectives

Focusing on designing and validating a practical OTA network layout suitable for implementation in a controllable environment[6]. The specific role of Software Defined Radios (SDRs) and MATLAB/simulation tools for prototype validation and addressing practical limitations such as synchronization and channel impairments [2], [6], [7].

0.c Thesis Contributions and Organization

The aim of this thesis is to contribute to the research objectives...as a student member of the university of western macedonia, department of electrical and computer engineering, under the supervision of professor ....

1 Ideal Over-the-Air Computation

Before introducing the practical impairments inherent to wireless channels, such as Inter-Symbol Interference (ISI), multipath fading and symbol to filter mismatch, it is essential to establish a baseline model of OTA computation under ideal conditions. In this scenario, the wireless medium is assumed to be perfectly synchronized, lacking frequency-selective fading, and corrupted only by Additive White Gaussian Noise (AWGN). This chapter aggregates the mathematical formulation, MATLAB code, Simulink models, and GNU Radio flowgraphs that demonstrate this ideal baseline.

1.a Mathematical Formulation of the Ideal Channel

Consider a wireless sensor network consisting of K End Device (ED) transmitting analog values to a Fusion Center (FC). Let $x_k(t)$ denote the continuous-time signal transmitted by the k-th device. In an ideal Multiple Access Channel (MAC) where perfect Channel State Information (CSI) allows for perfect pre-equalization of phase and amplitude shifts, the signals superimpose perfectly at the receiver.

The received baseband signal $y(t)$ at the FC is mathematically expressed as:

$$y(t) = \sum_{k=1}^{K} x_k(t) + n(t) \tag{1}$$

where $n(t) \sim \mathcal{N}(0, \sigma^2)$ represents the AWGN. Because there is no multipath delay spread, the channel impulse response is effectively a Dirac delta function $h(t) = \delta(t)$, meaning the current symbol is entirely unaffected by past or future symbols.

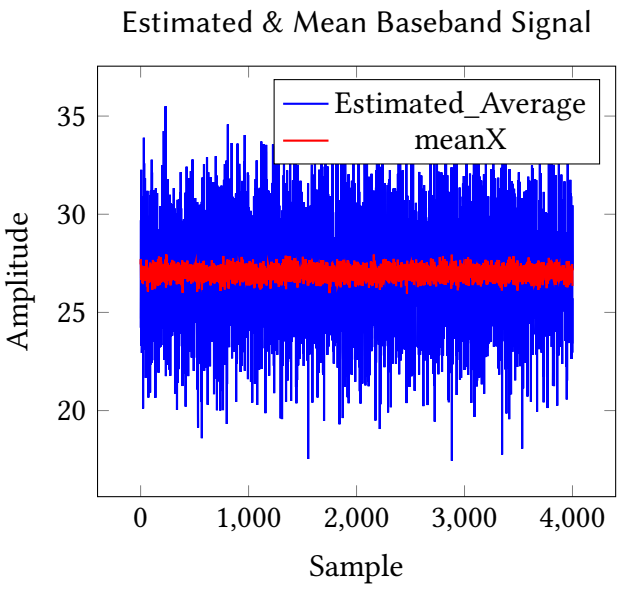
1.b MATLAB Simulation and Code Analysis

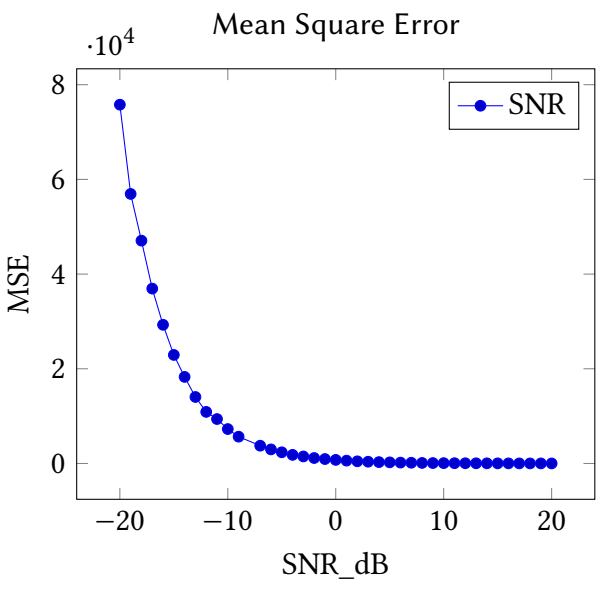
To validate this superposition principle, the ideal environment can be modeled using MATLAB. The following script generates normally distributed simulated sensor readings, aligns them perfectly in time, and adds AWGN to emulate the ideal MAC.

Listing 1: MATLAB script for ideal OTA computation aggregation.

```
% Ideal OTA Computation Simulation
K = 3;                    % Number of sensors
N = 1000;                 % Number of samples
SNR_dB = 20;              % Signal-to-Noise Ratio

% Generate random temperature readings around 27 degrees
X = 27 + randn(K, N);

% The Air acts as an ideal adder (MAC Superposition)
Ideal_Sum = sum(X, 1);

% Add AWGN
Received_Signal = awgn(Ideal_Sum, SNR_dB, 'measured');

% Calculate the Fusion Center Average
Estimated_Average = Received_Signal / K;

% Display Mean Squared Error (MSE)
MSE = immse(Estimated_Average, mean(X, 1)); fprintf('Ideal Baseline MSE
   at %d dB SNR: %f\n', SNR_dB, MSE);
```

As demonstrated by the script, when synchronization is perfect, the MSE is strictly a function of the AWGN variance.





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function received at the FC, denoted as $s[n]$, is the direct sum of the transmitted signals:

$$s[n] = \sum_{i=1}^{K} x_i[n] \tag{4}$$

where K is the total number of transmitting devices. For the given sensor data, the ideal aggregated sequence is evaluated as $s[n] = [3, 1, 0]$.

2.a.ii Multipath Channel Distortion

In practical agricultural deployments, the physical environment exhibits multipath characteristics (e.g., reflections from terrain or machinery). This is modeled as a Finite Impulse Response (FIR) filter. Assuming a channel impulse response $h[n]$ with a span of three taps, defined as $h[n] = [1.0, 0.5, -0.2]$, the actual received signal $y[n]$ is the convolution of the ideal aggregated signal with the channel response:

$$y[n] = s[n] * h[n] = \sum_{k=0}^{n} s[k]h[n-k] \tag{5}$$

Evaluating this convolution yields the distorted sequence $y[n] = [3.0, 2.5, -0.1, -0.2, 0.0]$. The extension of the sequence length from three to five symbols mathematically illustrates the presence of ISI, wherein previous and future symbols bleed into the current sampling instance.

2.a.iii Mean Squared Error (MSE) Evaluation

To quantify the degradation caused by the multipath channel, the MSE is calculated between the ideal target function $s[n]$ and the actual received signal $y[n]$. The error at each discrete time step is given by $e[n] = y[n] - s[n]$. For a sequence of length N, the overall MSE is defined as:

$$\text{MSE} = \frac{1}{N} \sum_{n=0}^{N-1} (y[n] - s[n])^2 \tag{6}$$

For the evaluated sequences, the discrepancy between the ideal sum and the ISI-corrupted signal results in an MSE $= 0.46$. This theoretical error establishes the baseline challenge that must be addressed through waveform design or digital equalization techniques at the FC.

2.b Practical Challenges

2.b.i Channel State Information (CSI) Imperfection

The theoretical model assumes perfect knowledge of the channel impulse response. In practice, the need for accurate and fresh CSI at transmitters or receivers is critical for precoding and equalization techniques [2], [6], [8]. Obsolete or noisy CSI severely limits the ability to pre-distort signals, exacerbating the MSE.

2.b.ii Waveform and Filter Design

As demonstrated by the sequence expansion in the multipath model, standard digital communication transceivers suffer from time sampling errors and ISI. Designing custom pulse shapes, such as optimized Root Raised Cosine (RRC) filters or DNN-generated waveforms, is essential to force zero-crossings at the exact sampling instances, thereby suppressing ISI natively over the air [2].

2.c Optimization Strategies to Enhance OAC Performance

To mitigate the aforementioned challenges, several optimization frameworks have been proposed in the literature:

- **Power Management**: Strategies to mitigate aggregation errors involve optimizing power control policies at the EDs to align received signal amplitudes, thereby minimizing the overall MSE or maximizing the convergence speed of federated learning algorithms [1], [2], [6].
- **Transceiver Design**: The joint design of pre-processing functions at the transmitter and post-processing equalizers at the receiver allows the system to match the channel characteristics to the desired nomographic function, effectively canceling multipath echoes and minimizing computational distortion [4], [5], [6], [8].

3 Experimental Methodology and Implementation

3.a Analog OAC and Nomographic Functions

To validate the mathematical models established in the previous section, an experimental framework was developed. The implementation focuses on analog OTA to compute nomographic functions—specifically, the arithmetic mean of distributed temperature sensors. By leveraging the superposition property of the multiple access channel, the network natively aggregates uncoded sensor readings in the air, transforming the physical medium into a computational processor.

3.b GNU Radio System Architecture

To physically simulate the analog OTA network, a Software Defined Radio (SDR) architecture was constructed using GNU Radio. The simulation environment accurately models the concurrent transmission of multiple Edge Devices (EDs) to a central Fusion Center (FC).

3.b.i Hardware and Software Parameters

The baseline configuration for the SDR implementation is detailed in Table 2. These parameters were selected to balance computational load while providing sufficient bandwidth to observe channel impairments.

3.b.ii Pulse Shaping and Channel Impairments

Two critical components of the GNU Radio flowgraph govern the physical behavior of the transmission:

- **Root Raised Cosine (RRC) Filter**: To constrain the transmission bandwidth and mitigate Intersymbol Interference (ISI), RRC filters are employed for pulse shaping at both the transmitters and the receiver. The RRC filter ensures that, under ideal Nyquist sampling rates, the overlapping tails of adjacent symbols evaluate to zero, thereby preserving the integrity of the target function computation.
- **Channel Model**: To accurately emulate an agricultural deployment, a Channel Model block is introduced to inject realistic wireless impairments. This block models Additive White Gaussian Noise (AWGN) and user-defined multipath delay taps, which scatter the signal arrivals and intentionally induce ISI to test the system's robustness.

3.b.iii Flowgraph Configurations

The network was simulated under two primary conditions. Figure 5 illustrates the continuous-time analog simulation where sensor data is continuously evaluated. Figure 6 depicts the "Faster-Than-Nyquist" (FTN) scenario, wherein the transmission rate is artificially accelerated to break the Nyquist limit and observe the resulting ISI degradation.

Table 2: Key SDR Hardware and Software Parameters for the OAC Implementation

Parameter	Value	Description
Hardware Configurations		
Center Frequency (f_c)	2.4 GHz	ISM band, selected to minimize external interference.
Sample Rate (f_s)	32 kHz	Defined baseband rate for the SDR sink/source blocks.
Antenna Configuration	SISO	Single-Input Single-Output for initial baseline testing.
Software & Baseband Processing		
Modulation Scheme	PAM	Pulse Amplitude Modulation for analog temperature transmission.
Samples per Symbol (sps)	6	Determines the interpolation/decimation rate for the signal.
Pulse Shaping Filter	RRC	Root Raised Cosine filter applied to mitigate ISI.
Roll-off Factor (α)	0.3	Standard excess bandwidth factor defining the filter's frequency transition band.

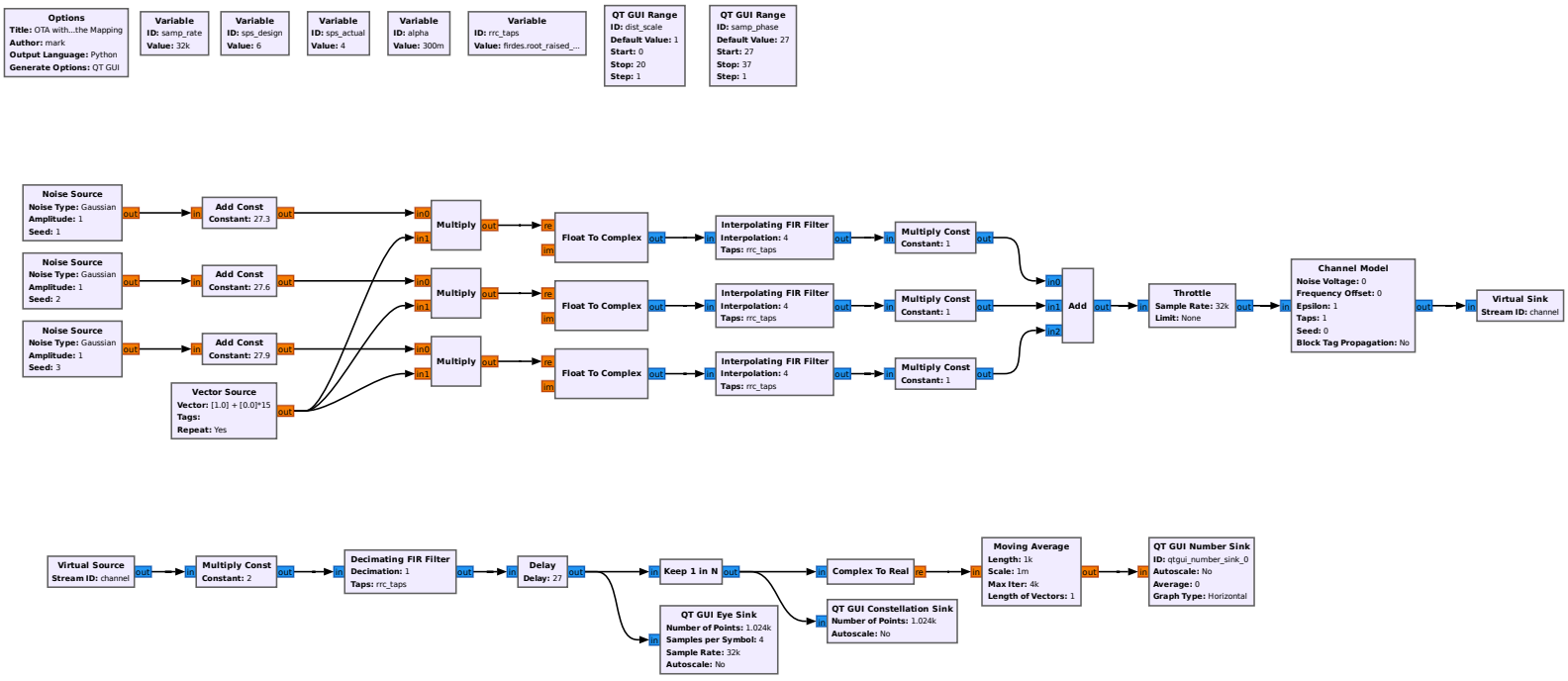

Figure 5: GNU Radio Companion flowgraph for the OTA implementation including Channel Model and RRC filters.

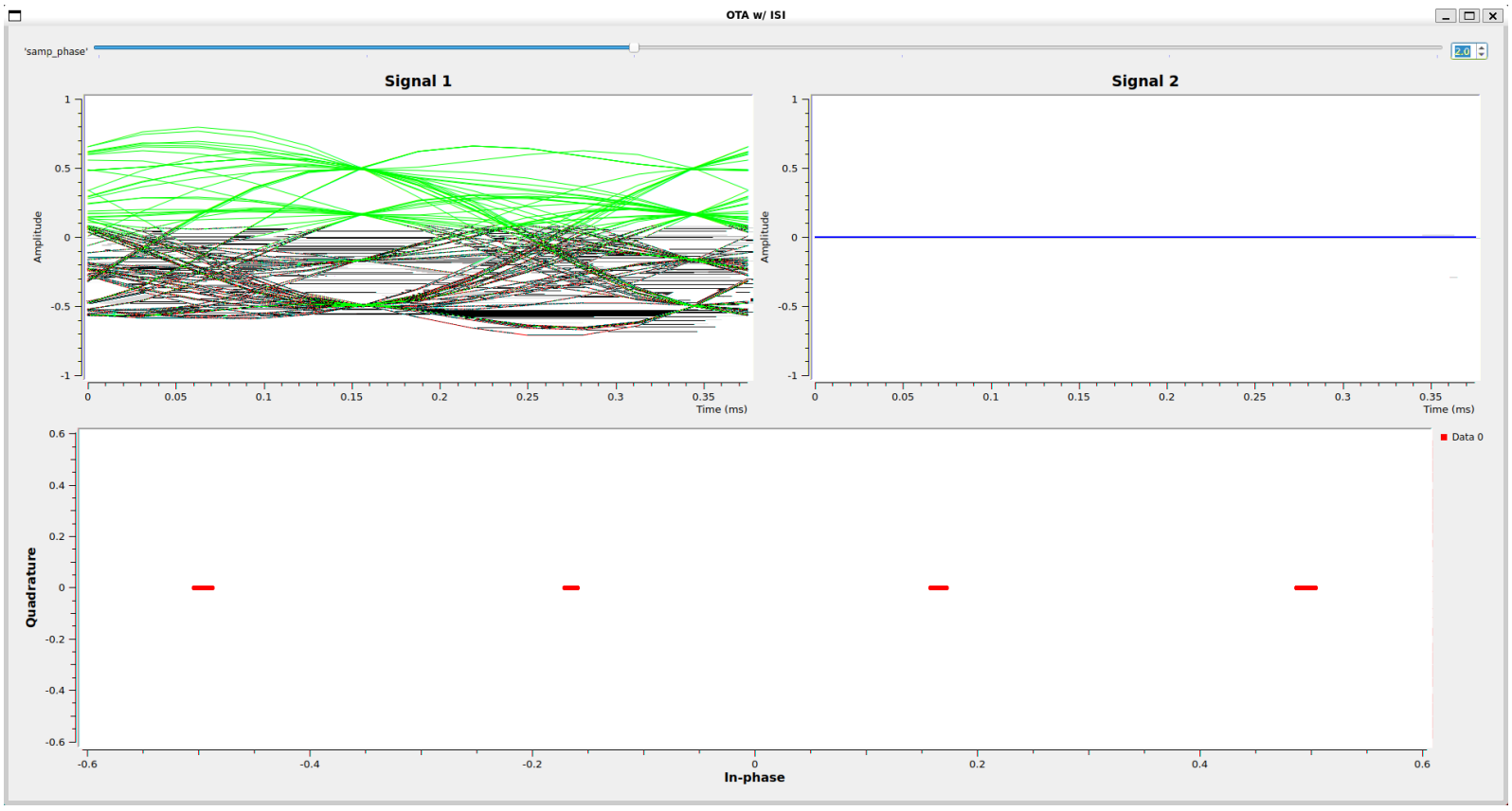

Figure 6: GNU Radio Companion flowgraph configured to induce Intersymbol Interference (ISI) by mismatching the design and actual sampling rates.

3.c MATLAB and Simulink Integration

While GNU Radio provides a robust framework for real-time SDR execution and analog signal evaluation, future phases of this methodology will incorporate MATLAB Simulink. Simulink will be utilized to model advanced feedback loops, custom waveform generation, and digital equalizers (e.g., Minimum Mean Square Error (MMSE) receivers) that are mathematically complex to implement in standard GNU Radio environments. This dual-software approach allows for both practical SDR validation and rigorous mathematical algorithmic design.

4 Experimental Results and Discussion

4.a Validation of the Implemented OAC Network

Demonstration of successful signal superposition and computation via the SDR platform[3], [6]. Comparison of achieved computation result against the ideal theoretical value[6], [9].

4.b Performance Evaluation under Impairments

Quantifying the performance metrics, such as Mean Squared Error (MSE) or Computation Error Rate (CER), of the implemented scheme[2], [6]. Analysis of the impact of deliberately introduced synchronization errors or varying channel noise (SNR) on the computation accuracy[2], [6]. Comparison with Separation Schemes (Simulated Baseline); Illustrating the resource savings and computation rate achieved by the OAC approach relative to traditional orthogonal access (e.g., Time-Division Multiple Access (TDMA))[3], [4], [6].

5 Conclusion and Future Work

5.a Summary of Thesis Findings

Recap of the feasibility and performance demonstrated by the SDR-based OAC network.

5.b Future Research Directions

Exploring more complex OAC architectures, such as multi-antenna systems Multiple-Input Multiple-Output (MIMO) or using Reconfigurable Intelligent Surfaces (RIS)[2], [10]. Further investigation into standardized OAC protocols for compatibility with modern wireless networks (e.g., LTE/5G protocols)[6], [9].

A MATLAB Code for OTA Simulation

This appendix contains the core MATLAB scripts used to generate the simulation data discussed in Chapter 2. Note that we utilized the Raised Cosine method [6].

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